Computational Evidence for Buddhist-Shakta Syncretism in Eastern India: Vocabulary Migration from Vajrayana Tara to the Bengali Baul Tradition

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Abstract

We present a computational study of vocabulary migration across six centuries of Bengali devotional literature, tracing the transmission of Buddhist Vajrayana terminology into Shakta Tantra and the Baul tradition of Bengal. Using a corpus of 48 texts across six tradition layers: Vajrayana Buddhist Tara texts (8th–12th c.), Old Bengali Charyapada (8th–12th c.), Bridge Tara texts (12th–16th c.), Shakta Kali and Durga texts (14th–18th c.), Bengali Shakta Padavali (18th c.), and Baul devotional songs (18th–19th c.): we apply TF-IDF vectorization with character n-gram features and cosine similarity analysis. Four principal findings emerge. First, three Tara texts from the Brihannilatantra tradition show Shakta-to-Buddhist vocabulary ratios of 2.0–4.0x, constituting measurable evidence of lexical migration from Buddhist to Shakta framing. Second, the Bridge Tara tradition as a cluster sits linguistically closer to Shakta Kali (cosine similarity 0.42) than to the Buddhist Tara texts from which it derives (0.31). Third, Charyapada-to-Baul similarity (0.40) is the strongest link in the entire corpus, showing that the Sahajiya Buddhist vernacular chain was the primary transmission mechanism into modern Bengali devotional culture, not the Sanskrit Buddhist tradition (Buddhist Tara to Baul = 0.02). Fourth, Ramprasad Sen's Shakta Padavali, despite being devoted to Kali, preserves substantial Buddhist residue (Shakta/Buddhist ratio = 3.0 on 39,000 tokens), indicating continued syncretic vocabulary even within the mature Shakta tradition. These results provide the first quantitative corroboration of historically argued Buddhist-Shakta syncretism in Bengal using computational text analysis.

Keywords: computational humanities, Sanskrit NLP, Bengali devotional literature, Buddhist-Shakta syncretism, Tara, Charyapada, Baul tradition, TF-IDF, cosine similarity, cultural transmission

1. Introduction

The question of Buddhist influence on the Shakta Tantra tradition of eastern India has long occupied scholars of Indian religious history. Agehananda Bharati, Shashibhushan Dasgupta, and David Lorenzen have argued on philological and iconographic grounds that the goddess Tara, the Mahavidya Kali, and elements of Shakta ritual derive substantially from late Vajrayana Buddhist practice centered at the great monasteries of Bengal and Bihar, including Nalanda, Vikramashila, and Odantapuri, during the Pala dynasty (8th–12th centuries CE). After the collapse of the Pala state and the destruction of these centers, it has been argued that Buddhist practitioners and their vocabulary were absorbed into local Shakta and Sahajiya traditions, going 'underground' in the vernacular devotional poetry of the Charyapada and later re-emerging in the Baul traditions of modern Bengal.

This argument has remained largely qualitative. Scholars have pointed to shared iconography, overlapping epithets, and textual borrowings as evidence, but the claim has lacked systematic quantitative support. The present study addresses this gap by applying corpus analysis methods to a multi-script, multi-tradition text collection spanning the full historical arc from Sanskrit Vajrayana texts to modern Bengali Baul songs.

The specific questions we address are: (1) Can vocabulary migration from Buddhist to Shakta framing be measured quantitatively in texts nominally devoted to the same goddess (Tara)? (2) Does the cluster of Bridge Tara texts, i.e. texts presenting the Buddhist Tara in Shakta ritual framing, occupy a measurable intermediate position between Buddhist and Shakta traditions? (3) Which transmission channel was primary for the survival of Buddhist vocabulary into modern Bengali devotional culture: the Sanskrit textual tradition or the vernacular Sahajiya-Baul chain? (4) Does Ramprasad Sen's 18th-century Bengali Shakta Padavali preserve measurable Buddhist residue?

2. Corpus and Data

2.1 Corpus Design

We assembled a corpus of 48 text documents across six tradition layers, following a deliberate chain structure from the earliest Vajrayana Buddhist sources to the latest modern Bengali devotional texts. The six layers are: (1) Pure Vajrayana Buddhist Tara texts in Sanskrit and IAST transliteration, including the 21 Praises to Tara, the Sadhanamala-derived sadhanas, Hevajra Tantra excerpts, and related Vajrayogini and Vajradevi stotras; (2) The Charyapada, Old Bengali doha verses (8th–12th c.) attributed to the 84 Mahasiddhas; (3) Bridge Tara texts presenting the Buddhist goddess Tara in explicitly Shakta framing, primarily from the Brihannilatantra; (4) Shakta Kali texts including two independent Kali sahasranamas and related stotras; (5) Shakta Durga texts including the Devi Mahatmyam and the Mahishasura Mardini namavali; (6) Bengali Shakta Padavali (Ramprasad Sen, 18th c.) and Baul-Bengali devotional texts (Lalon Fakir, Tagore's Gitanjali).

2.2 Data Sources and Processing

Sanskrit and Devanagari texts were sourced primarily from sanskritdocuments.org and GRETIL. Bengali texts were sourced from archive.org (Ramprasad Sen), lalalgeeti.com (Lalon Fakir songs), and the Society for Natural Language Technology Research edition of Tagore's Gitanjali. The Ramprasad Sen Kabya Sangraha (372 pages, scanned image PDF) was processed using Tesseract OCR 5.3.4 with the Bengali language model at 300 DPI, yielding 39,067 clean Bengali tokens. The Charyapada corpus was sourced as a digitized Bengali Unicode text.

Text files exist in three scripts: Devanagari Unicode (Sanskrit texts), Bengali Unicode (Charyapada, Ramprasad, Lalon, Gitanjali), and IAST/Roman transliteration (Hevajra, Devi Mahatmyam, Aryamanjushrimula, some Tara texts). Cross-script comparisons are methodologically constrained: the character n-gram vectorizer correctly identifies orthographic similarity within a script but cannot bridge script boundaries, which we discuss in Section 5.

3. Methodology

3.1 Vectorization

We represent each document as a TF-IDF vector using character n-grams of length 3 to 5. Character n-grams are appropriate for Sanskrit and Bengali for three reasons. First, Sanskrit is a highly inflected language where the same root appears in many morphological forms; character n-grams capture shared root syllables (e.g., tar-, karun-, bhay-, vajr-) without requiring a morphological analyzer. Second, the same method applies across scripts without modification, making cross-tradition comparisons within a single script consistent. Third, character n-grams are robust to OCR noise, which is a practical advantage for the Ramprasad material. We use sublinear TF scaling and a vocabulary ceiling of 25,000 features.

3.2 Similarity Metric

Pairwise cosine similarity is computed between all document vectors. Cross-tradition averages are computed as the mean cosine similarity between all pairs (i, j) where document i belongs to tradition A and document j belongs to tradition B and $i \neq j$. All tradition-level similarities reported in Section 4 use these averages. For the Devanagari-only subset analysis (Table 3), we restrict to documents in Devanagari script to eliminate cross-script orthographic noise.

3.3 Term Density Analysis

In parallel with the n-gram similarity analysis, we count occurrences of curated Buddhist and Shakta term lists directly in the text of each document, normalizing by document length (per 1,000 tokens). The Buddhist term list includes: बुद्ध, बोधिसत्त्व, बोधि, करुणा, प्रज्ञा, शून्य, तथागत, वज्र, नैरात्म, सहज, महासुख, योगिनी, डाकिनी, हेवज्र, तारा, चर्या (Devanagari), and their Bengali Unicode equivalents. The Shakta term list includes: शक्ति, शिव, महादेव, पार्वती, दुर्गा, काली, महाविद्या, तन्त्र, मन्त्र, यन्त्र, कुल, श्रीविद्या, भैरव. Term density ratios (Shakta/Buddhist per 1,000 tokens) are used to characterize the vocabulary orientation of individual texts and tradition clusters.

4. Results

4.1 Cross-tradition Cosine Similarities

Table 1 presents the full cross-tradition similarity matrix. The central finding is the ordering of inter-tradition similarities, which traces two distinct transmission channels.

Table 1: Cross-tradition cosine similarity (character 3–5 gram TF-IDF, full corpus)

Tradition Pair	**Avg Cosine Similarity**	**n (docs)**	**Interpretation**
Buddhist Tara → Bridge Tara	0.3095	11 × 11	Sanskrit Buddhist → Shakta Tara
Bridge Tara → Shakta Kali	0.4197	11 × 7	Highest Sanskrit link; bridge fully Shakta-ward
Buddhist Tara → Shakta Kali	0.3257	11 × 7	Direct Buddhist → Kali (weaker than via bridge)
Buddhist Tara → Shakta Durga	0.0948	11 × 5	Low: different textual genre (narrative vs. stotra)
Kali → Durga (within Shakta)	0.0524	7 × 5	Genre gap within Shakta tradition
Shakta Kali → Shakta Padavali	0.0000	7 × 1	Script boundary: Sanskrit→Bengali (methodological)
Shakta Padavali → Baul	0.4060	1 × 3	Strong Bengali vernacular chain: Ramprasad → Lalon/Tagore
Charyapada → Baul	0.4022	2 × 3	Strongest vernacular link: Old Bengali Buddhist → modern

Tradition Pair	Avg Cosine Similarity	n (docs)	Interpretation
Buddhist Tara → Baul	0.0181	11 × 3	Near-zero: Sanskrit Buddhist leaves no direct Baul trace

The Sanskrit channel shows a monotonically increasing similarity from Buddhist to Bridge to Shakta: Buddhist Tara to Bridge Tara (0.31), Bridge Tara to Shakta Kali (0.42). Crucially, the Bridge Tara tradition is measurably closer to Shakta Kali than to the Buddhist Tara texts from which it derives. This 'crossing over', i.e. a higher similarity to the destination tradition than to the source, constitutes the primary evidence for completed lexical migration.

The vernacular Bengali channel shows equally strong links: Shakta Padavali to Baul (0.41), Charyapada to Baul (0.40). The near-zero similarity of Sanskrit Buddhist Tara to Baul (0.02) confirms that the Sanskrit tradition left essentially no direct orthographic trace in modern Bengali devotional vocabulary. The primary transmission mechanism was the vernacular Sahajiya-Buddhist chain, not the Sanskrit textual tradition.

4.2 Vocabulary Migration in Bridge Tara Texts

Table 2 presents term density analysis for individual Bridge Tara texts, showing the Buddhist-to-Shakta vocabulary ratio per document.

Table 2: Vocabulary orientation of Bridge Tara texts (normalized per 1,000 tokens)

Text	Tokens	Buddhist/1k	Shakta/1k	S/B Ratio	Status
tArAsahasranamastotra (Brihannilatantra)	1,710	2.34	7.02	3.0	MIGRATED
Tara Stotram / Nila Sarasvati Ashtakam	645	1.55	4.65	3.0	MIGRATED
Tarashatanamastotra (Brihannilatantra)	492	0.00	4.07	4.0+	MIGRATED
Shri Tara Sahasranama Stotram 3	1,625	5.54	5.54	1.0	At parity
Shri Tara Takaradi Sahasranama	1,342	2.24	0.00	0.0	Buddhist-dominant
Shrimad Ugratara Hridayastotram	354	5.65	0.00	0.0	Buddhist-dominant

Three texts show Shakta/Buddhist ratios of 2.0 or greater, flagged as 'migrated': the tArAsahasranamastotra from the Brihannilatantra (ratio 3.0, 1,710 tokens), the Tara Stotram / Nila Sarasvati Ashtakam (ratio 3.0, 645 tokens), and the Tarashatanamastotra also from the Brihannilatantra (ratio 4.0+, 492 tokens). These three texts are nominally Tara texts but use Shakta vocabulary at three to four times the density of Buddhist vocabulary. Importantly, these three migrated texts are all associated with the Brihannilatantra, a text known to represent the Shakta absorption of Buddhist Tara. The Takaradi Sahasranama, by contrast, which derives from the Brahmayamala, retains Buddhist-dominant vocabulary (ratio 0.0). This text-source correlation supports the hypothesis that migration was not uniform across the Tara textual tradition but concentrated in specific tantric lineages.

4.3 Tradition-level Vocabulary Gradient

Table 3 presents normalized term densities aggregated at the tradition level, showing a gradient in vocabulary orientation from the full corpus.

Table 3: Tradition-level vocabulary orientation (Devanagari and Bengali texts, per 1,000 tokens)

Tradition	Avg Buddhist/1k	Avg Shakta/1k	S/B Ratio	Script
Buddhist Tara	4.22	1.18	0.3	Devanagari
Bridge Tara	3.56	3.22	0.9	Devanagari
Shakta Kali	3.83	5.80	1.5	Devanagari
Shakta Padavali (Ramprasad)	2.89	8.78	3.0	Bengali
Baul/Bengali (Lalon, Tagore)	1.12	0.25	0.2	Bengali
Charyapada	7.33	1.61	0.2	Bengali

The Buddhist/Shakta ratio shows a clear monotonic progression across the Sanskrit chain: Buddhist Tara (0.3) → Bridge Tara (0.9) → Shakta Kali (1.5), and then a sharp jump in the Bengali Shakta Padavali (Ramprasad: 3.0). The Bridge Tara cluster sits at near-parity (0.9), confirming its intermediate character. The Charyapada, despite being a Buddhist corpus, shows a high Buddhist density as expected (ratio 0.2 from the Buddhist side, i.e., Buddhist >> Shakta), providing the calibration anchor for the vernacular chain.

4.4 The Ramprasad Finding

Ramprasad Sen's Kabya Sangraha presents a striking result: despite being a corpus of Shakta Kali devotional songs (কালী: 103 occurrences, শ্যামা: 29, তারা: 56, মা: 196), it retains 113 Buddhist-coded term hits across 39,067 tokens. The word তারা (Tara) appears 56 times, compared to কালী 103 times. Ramprasad addresses the goddess variously as Tara, Kali, Shyama, and Ma within the same song corpus, treating these as interchangeable names. The Shakta/Buddhist ratio of 3.0 on this large corpus is statistically stable, and the Buddhist residue (2.89 per 1,000 tokens) is higher than in the Bridge Tara cluster (3.56 per 1,000 tokens). This indicates that the mature 18th-century Bengali Shakta tradition actively preserved Buddhist vocabulary as part of its devotional register, not merely as etymological residue.

5. Discussion

5.1 Two Transmission Channels

Our results support a model of two distinct but parallel transmission channels for Buddhist vocabulary into modern Bengali devotional culture. The first is the Sanskrit channel: Vajrayana Buddhist Tara texts → Bridge Tara (Brihannilatantra tradition) → Shakta Kali sahasranamas. This channel operated primarily through Sanskrit textual composition and is visible in the high inter-tradition cosine similarities within the Devanagari corpus (0.31–0.42). The second is the vernacular Bengali channel: Charyapada Sahajiya Buddhism → Shakta Padavali → Baul tradition. This channel operated through oral transmission and vernacular poetic composition and is visible in the strong Bengali-to-Bengali similarities (0.40–0.41).

The near-zero similarity between Sanskrit Buddhist Tara and the Baul tradition (0.02) shows that these two channels did not directly communicate at the orthographic level. Sanskrit Buddhist vocabulary did not flow directly into Baul Bengali. The Baul vocabulary is a transformation of Sahajiya Buddhist vocabulary as mediated through the Old Bengali Charyapada tradition, not of the Sanskrit Vajrayana corpus. This is consistent with the historical account in which the Pala monasteries were destroyed (12th–13th c. CE) while the Sahajiya tradition survived in rural Bengal through the Baul lineage.

5.2 The Script Boundary as Evidence

The zero similarity between Sanskrit Shakta Kali texts and Bengali Shakta Padavali (Ramprasad) is methodologically expected, since character n-grams cannot bridge the Devanagari/Bengali orthographic boundary, but it is also historically meaningful. It shows that the transmission from Sanskrit Shakta ritual tradition to Bengali Shakta devotional poetry did not happen through direct textual borrowing at the lexical level. Ramprasad did not copy Sanskrit Kali sahasranama vocabulary into his Bengali songs. Instead, both traditions drew on a shared pool of Shakta devotional concepts that were encoded independently in their respective scripts and registers. The script boundary in the similarity matrix is not a measurement failure; it is a correct reflection of an oral-to-written transmission that left no orthographic fingerprint.

5.3 Limitations

Several limitations of the study should be noted. The term lists used for Buddhist and Shakta vocabulary are necessarily selective; a more comprehensive Sanskrit and Bengali lexical resource would improve term density analysis. The character n-gram approach, while effective for capturing shared root morphology, treats Devanagari and Bengali as orthographically distinct, which prevents direct cross-script similarity computation. The Ramprasad corpus was obtained through OCR from a scanned 1914 edition and contains approximately 5–10% character-level noise, which may slightly affect token counts but is unlikely to change tradition-level term density ratios significantly. Finally, our tradition classifications are based on textual source attribution, not independent philological dating; future work should incorporate manuscript dating and colophon evidence.

6. Conclusion

We have presented the first computational corpus analysis of Buddhist-Shakta textual syncretism in Bengal, spanning six tradition layers from 8th-century Sanskrit Vajrayana texts to 19th-century Bengali Baul songs. Four findings stand out. First, the Bridge Tara tradition as a whole has linguistically migrated Shakta-ward, with three specific texts (all from the Brihannilatantra) showing Shakta/Buddhist vocabulary ratios of 2.0–4.0x. Second, the tradition-level gradient from Buddhist Tara (ratio 0.3) through Bridge Tara (ratio 0.9) to Shakta Kali (ratio 1.5) shows a smooth, measurable vocabulary migration along the Sanskrit channel. Third, the Charyapada-to-Baul link (cosine 0.40) equals or exceeds the Shakta Padavali-to-Baul link (0.41), confirming that the primary vehicle for Buddhist vocabulary survival in modern Bengali devotional culture was the vernacular Sahajiya chain, not the Sanskrit tradition. Fourth, Ramprasad Sen's Bengali Kali songs preserve substantial Buddhist vocabulary residue, with Tara appearing 56 times in a corpus where Kali appears 103 times.

These results provide quantitative grounding for hypotheses about Buddhist-Shakta syncretism that have previously rested on philological argument and anecdotal textual evidence. The computational approach enables claims about the relative degree of vocabulary migration across different textual lineages, which philological analysis alone cannot easily provide.

The code and corpus used in this study are available at: https://github.com/joyboseroy/bengal-dharma-corpus

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Appendix A: Corpus Document List

The following 48 documents constitute the full corpus used in this study. All files are available in the project repository.

Document	Tradition Layer	Script	Tokens
Aryataranamaskaraikavimshati Stotra	Buddhist Tara	Devanagari	201
Arya Tara Stutih	Buddhist Tara	Devanagari	95
Aryatarasragdhara Stotram	Buddhist Tara	Devanagari	447
Aryatara Ashtottarashata Nama Stotra	Buddhist Tara	Devanagari	474
Shri Tara Dhyanam	Buddhist Tara	Devanagari	103
Prajnaparamita Stotram	Buddhist Tara	Devanagari	187
Nairatmashtaka Stotram	Buddhist Tara	Devanagari	64
Vajradevi Stotram	Buddhist Tara	Devanagari	170
Vajrayoginipranamaikavishika	Buddhist Tara	Devanagari	152
Vajrayoginyah Pindartha Stuti	Buddhist Tara	Devanagari	157
Vasudhara Stotram	Buddhist Tara	Latin/IAST	74
Hevajra Tantra (excerpts)	Buddhist Tara	Latin/IAST	1,415
Hevajra Sadhana	Buddhist Tara	Latin/IAST	363
Charyapada (47 padas)	Charyapada	Bengali	45,998
The Charypada (PDF scholarly ed.)	Charyapada	Latin	2,794
Shri Nila Sarasvati Stotram	Bridge Tara	Latin/IAST	229
Tara Stotram / Nila Sarasvati Ashtakam	Bridge Tara	Devanagari	645
tArAkavacham	Bridge Tara	Devanagari	631
tArAsahasranamastotra (Brihannilatantra)	Bridge Tara	Devanagari	1,710
Shri Tara Sahasranama Stotram 3	Bridge Tara	Devanagari	1,625
Shri Tara Shatanama Stotram	Bridge Tara	Devanagari	198
Shri Tara Takaradi Sahasranamastotra	Bridge Tara	Devanagari	1,342
Shri Tarashatanama Stotram	Bridge Tara	Devanagari	171
Tarashatanamastotra (Brihannilatantra)	Bridge Tara	Devanagari	492
Mahogratara Stuti	Bridge Tara	Devanagari	171
Mahogratarashtaka Stotram	Bridge Tara	Devanagari	169
Shrimad Ugratara Hridayastotram	Bridge Tara	Devanagari	354
Shri Kali Sahasranama Stotram (Kalikulasarvasva)	Shakta Kali	Devanagari	1,608
Shri Kali Sahasranama Stotram (Brihannilatantra)	Shakta Kali	Devanagari	1,608
Dakshinakalika Sahasranama Stotram	Shakta Kali	Devanagari	1,076
Shri Kali Shatanama Stotram	Shakta Kali	Devanagari	201
Shri Kali Tandava Stotram	Shakta Kali	Devanagari	72
Kali Stava Brahmakritam	Shakta Kali	Devanagari	113
Dakshaprokta Kali Stuti	Shakta Kali	Devanagari	174

Document	Tradition Layer	Script	Tokens
Devi Mahatmyam / Durga Saptashati	Shakta Durga	Latin/IAST	6,830
Markandeya Purana chs. 1–93	Shakta Durga	Latin/IAST	51,900
Shri Mahishasuramardini Ashtottarashata	Shakta Durga	Devanagari	324
DurgaStutiH (Essence of Devi Mahatmyam)	Shakta Durga	Latin/IAST	827
Part of Bhagavati Padyapushpanjali	Shakta Durga	Latin/IAST	1,566
Ramprasad Sen: Kabya Sangraha (OCR)	Shakta Padavali	Bengali	39,067
Lalon Fakir songs	Baul Bengali	Bengali	48,372
Gitanjali (Tagore, Bengali original)	Baul Bengali	Bengali	16,618